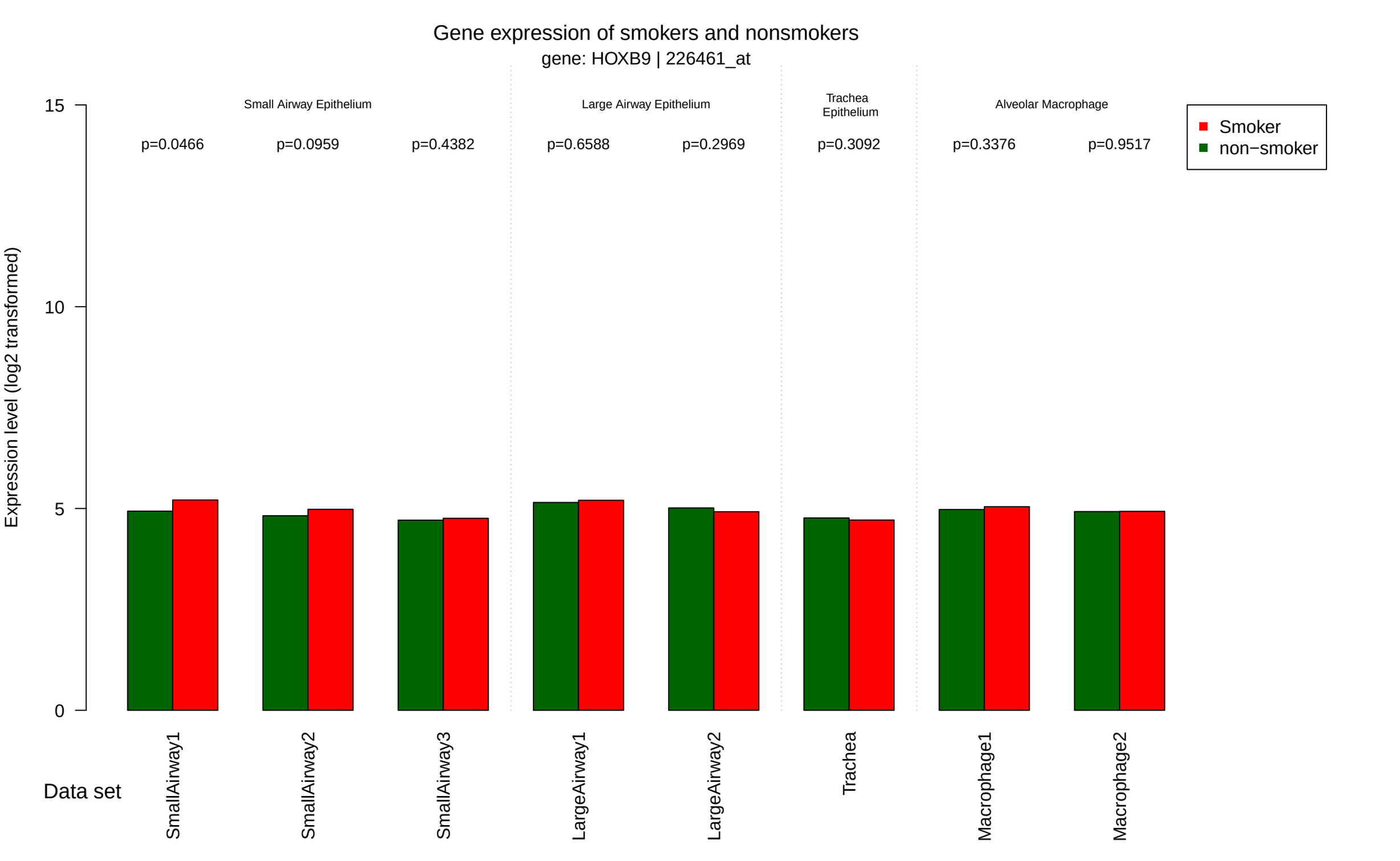
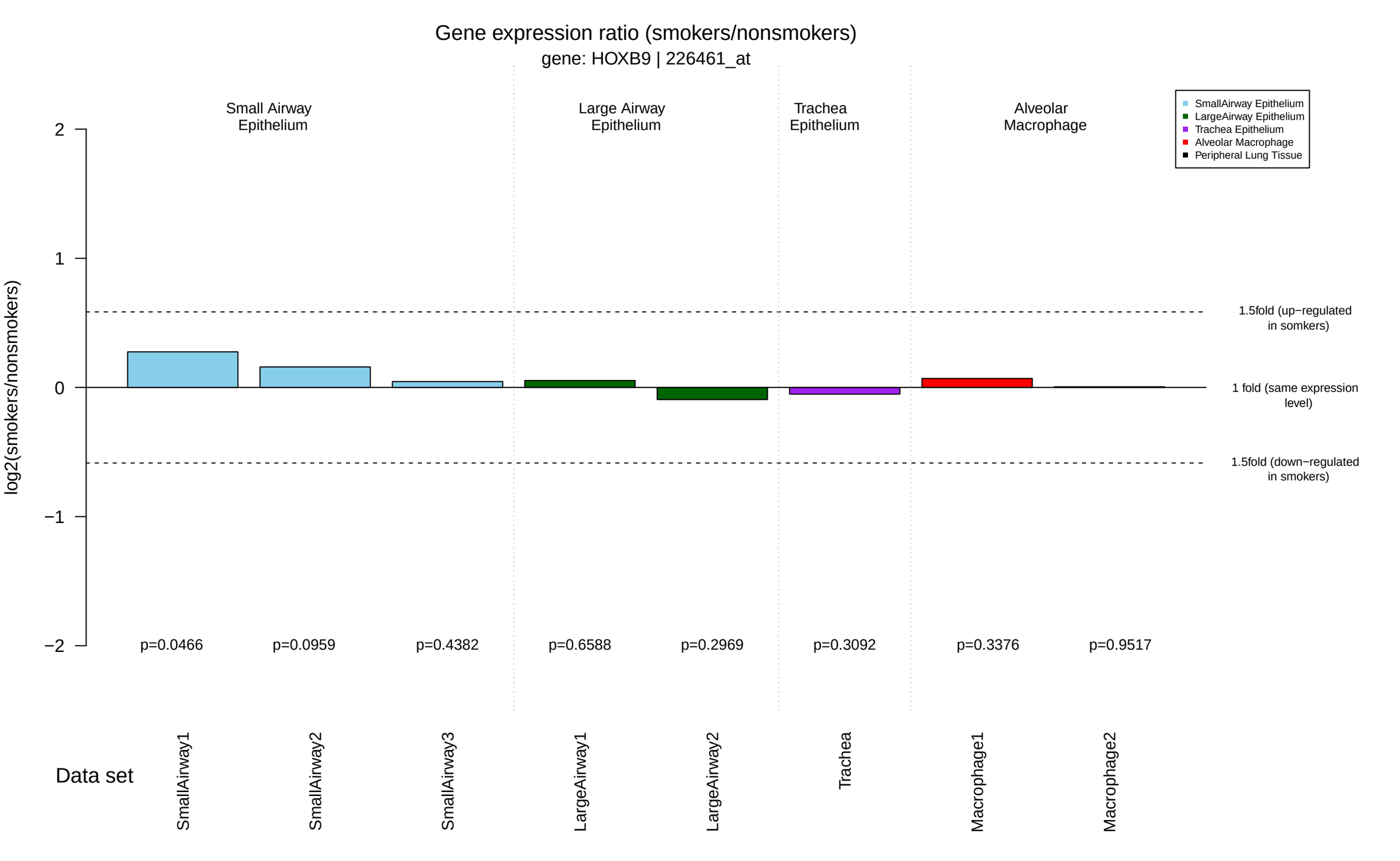
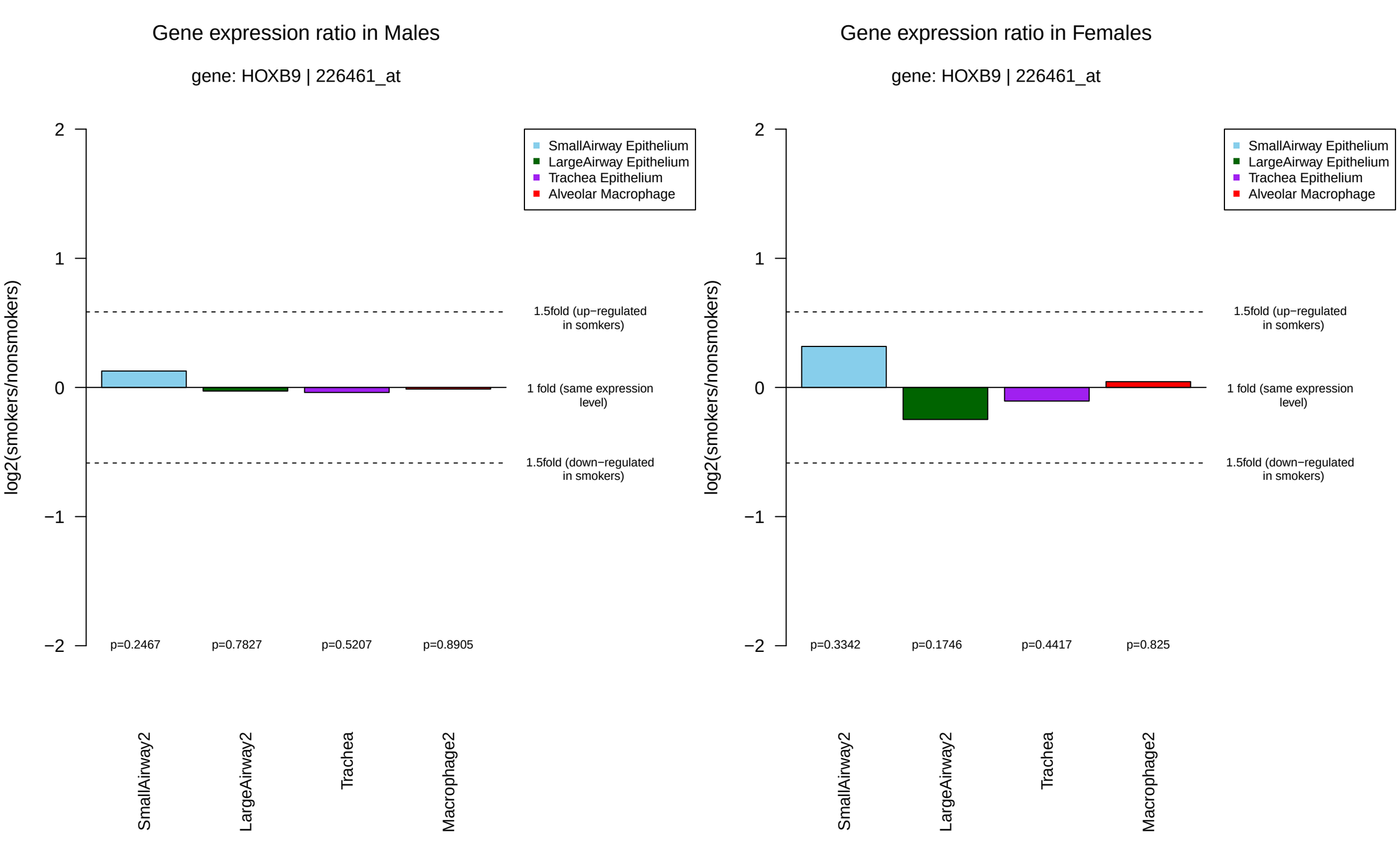
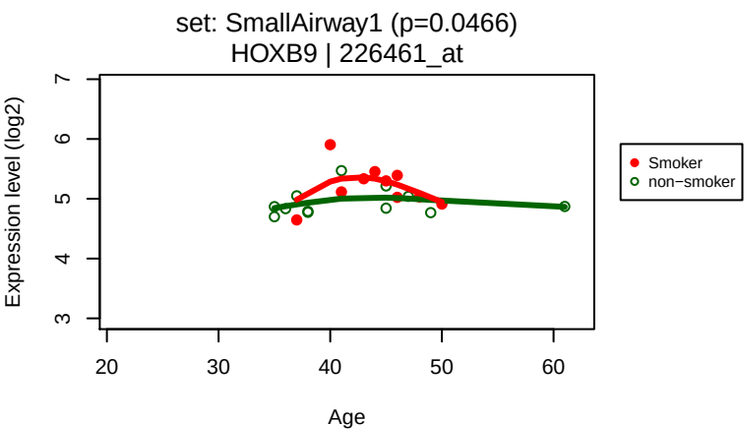
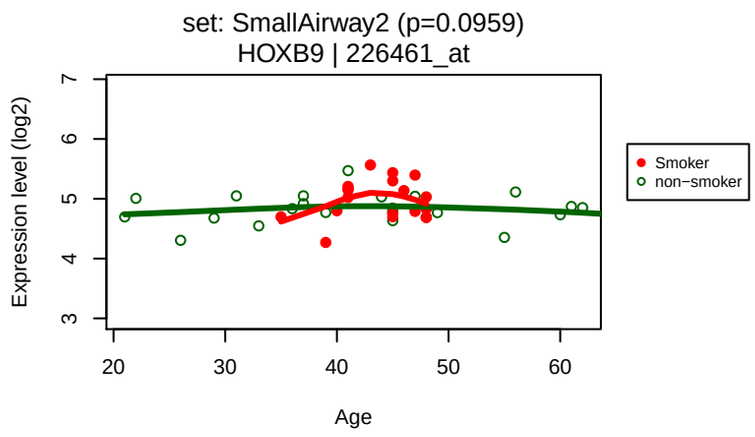
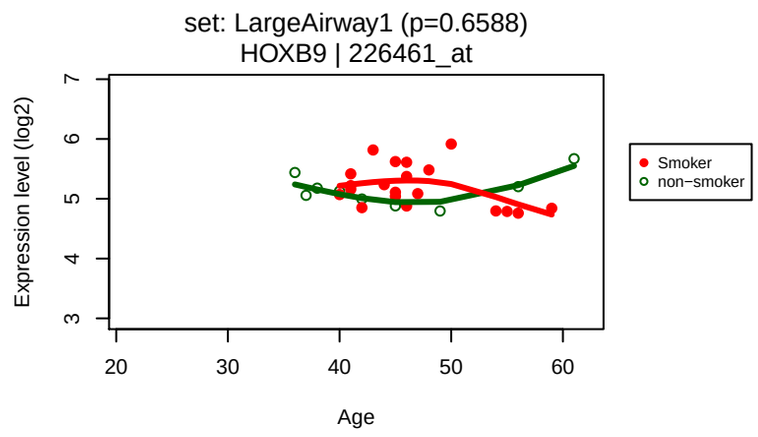

Gene expression of smokers and nonsmokers
gene: HOXB9 | 226461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0466
p=0.0959
p=0.4382
p=0.6588
p=0.2969
p=0.3092
p=0.3376
p=0.9517
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

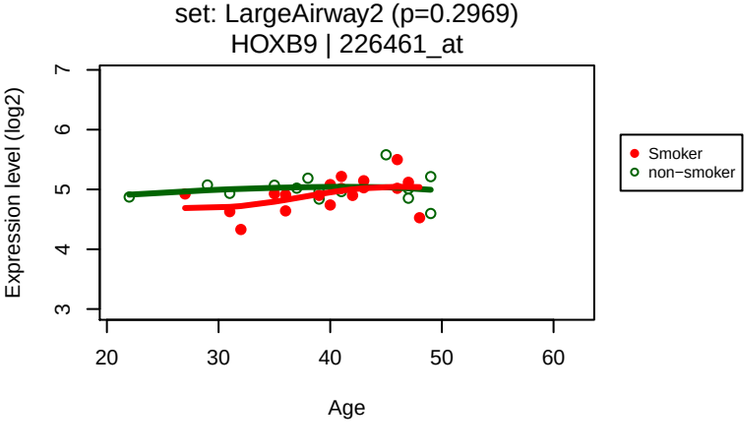

Gene expression ratio (smokers/nonsmokers)
gene: HOXB9 | 226461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0466
p=0.0959
p=0.4382
p=0.6588
p=0.2969
p=0.3092
p=0.3376
p=0.9517
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

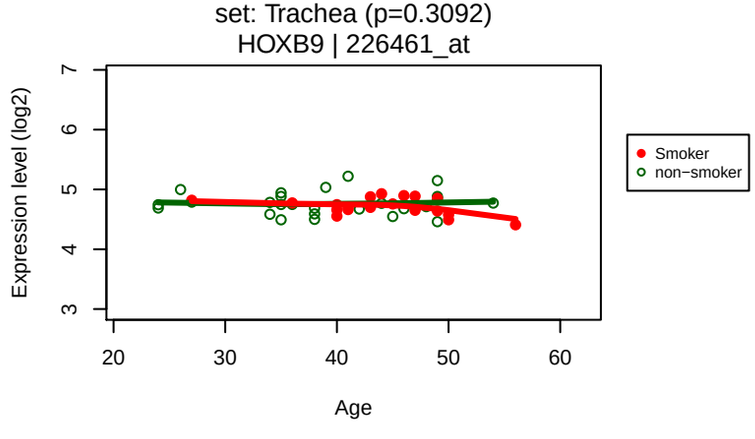

Gene expression ratio in Males
gene: HOXB9 | 226461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2467
p=0.7827
p=0.5207
p=0.8905
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: HOXB9 | 226461_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3342
p=0.1746
p=0.4417
p=0.825
SmallAirway2
LargeAirway2
Trachea
Macrophage2

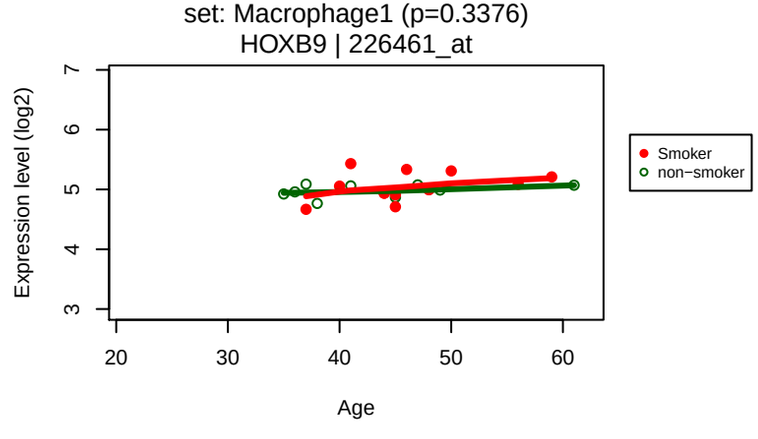

set: SmallAirway1 (p=0.0466)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker

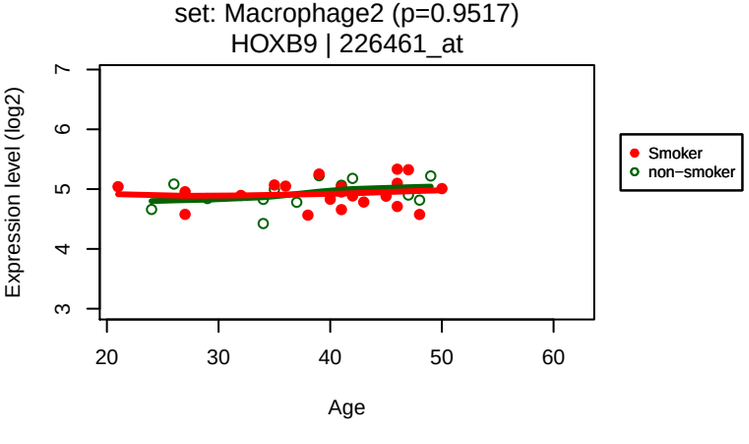

set: SmallAirway2 (p=0.0959)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker

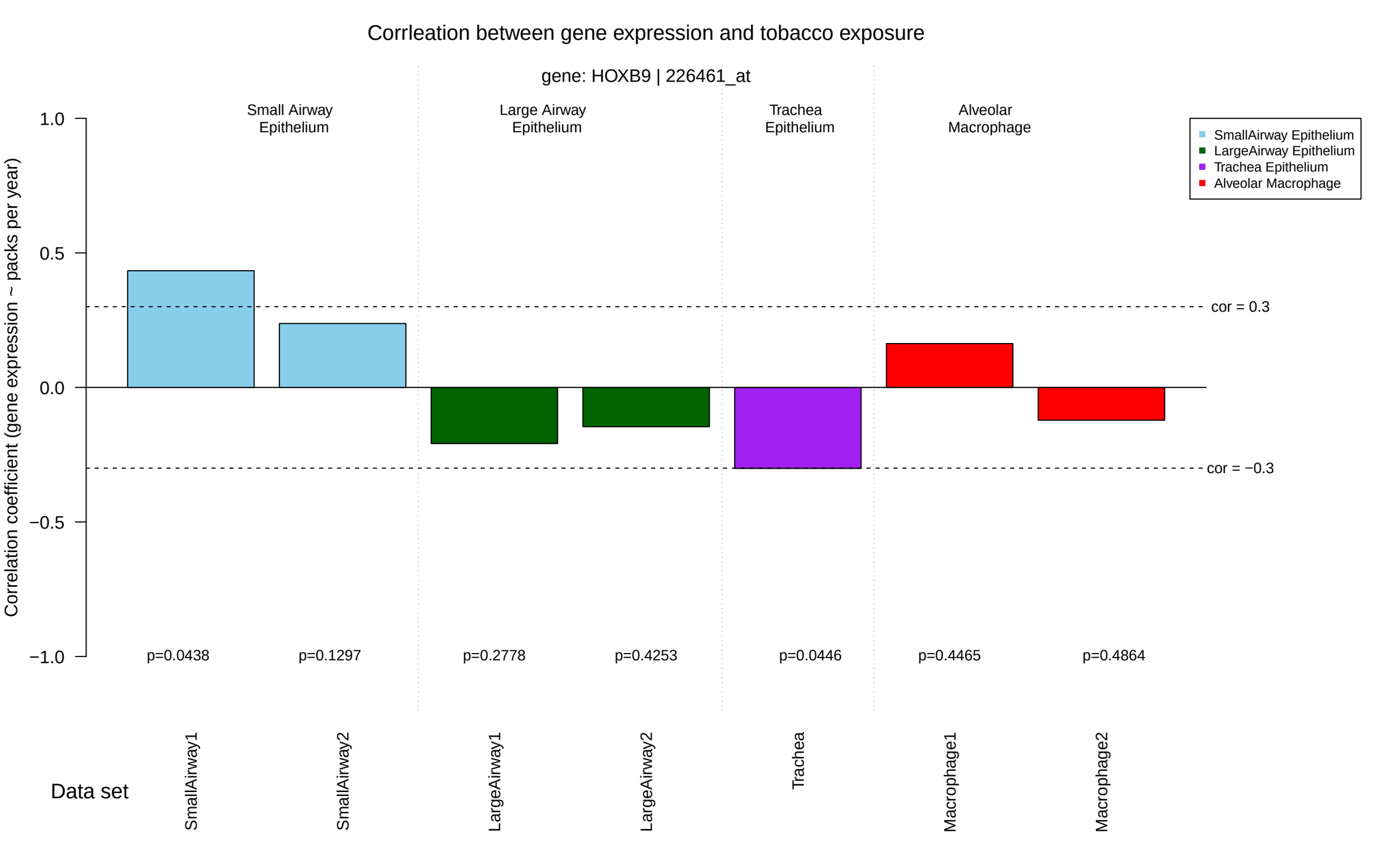

set: LargeAirway1 (p=0.6588)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2969)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3092)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3376)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.9517)
HOXB9 | 226461_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: HOXB9 | 226461_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0438
p=0.1297
p=0.2778
p=0.4253
p=0.0446
p=0.4465
p=0.4864
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2